

The Two-Round Blockholder Disclosure Model

A standalone model note

v4 theory lane, ticket 30 (T2j)

Card stamp 2026-08-21, post-ledger regeneration

Abstract

This note states the v4 two-round model in one place: primitives, timing, the partition the disclosure rule induces, the equilibrium notion, the standing hypotheses, and the eight-result ledger with its honesty labels. It then records what the v4 model takes from the frozen manuscript `draft_v2.tex`, what it simplifies, and what it drops; and it lists the executed evidence, failures included. Every statement of content here is transcribed from `research/model_v4/MODEL_CARD.md` (version stamp 2026-08-21, post-ledger regeneration) and `research/model_v4/LABEL_LEDGER.md`. The card is the single source of truth: where a proof file and the card differ, the card wins. The conditionality attached to each result is not decoration. It is the result.

1 Position and object

The disclosure rule *is* the market's partition. The model asks how liquidity κ moves bidder entry and the expected takeover premium when a stake threshold τ and a filing window T split control-node histories into a **flagged** cell (the filing has landed before the control decision) and a **pooled** cell (it has not).

The control-outcome object is the expected engagement-related premium $\Delta^{\text{act}}(\kappa, \tau, T)$. The price-path objects are the run-up path R_d , the cumulative run-up R , and the filing-day jump J . Lower τ means a tighter threshold margin; lower T means a tighter window margin.

Throughout, $\kappa \in [0, 1]$ is *noise-trading intensity* — never depth, volume or turnover. Upright T is the filing window; $\mathcal{T}$ is the outer best-response map. Bare λ is not available (it is the appropriability coefficient of the tender-game appendix), and neither is a bare C , W , $\mathcal{S}$, u or g : every one of those letters is overloaded in this project and always carries its subscript.

2 Primitives

Fundamentals, signal, bidder

- $v \sim N(\mu_v, \sigma_v^2)$ is the target's standalone value; the blockholder observes $s = v + \varepsilon$ with $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ independent of v . The Gaussian projection coefficient is $\beta = \sigma_v^2 / (\sigma_v^2 + \sigma_\varepsilon^2) \in (0, 1)$, so that $\mathbb{E}[v \mid s] = \mu_v + \beta(s - \mu_v)$. (The letter is β , the frozen manuscript's name; bare λ is reserved.)
- The bidder draws a private synergy shock $\xi \sim N(0, \sigma_\xi^2)$ independent of (v, s) ; $\bar{S}$ is mean bidder synergy and $K > 0$ the entry cost.
- m_0, m_1 are the takeover premia without and with engagement, with $m_1 > m_0$ and $m_0 \geq 0$; $\Delta_m = m_1 - m_0 > 0$ is the premium wedge, and $\Delta_V \geq 0$ is the non-takeover value engagement creates. The restriction $m_0 \geq 0$ is load-bearing: it is what makes the inner pricing fixed point exist, be unique and be continuous (Section 6, A5).

Trading, stakes, the legal clock

- Noise is a ternary mark of size $\bar{z} > 0$: $\mathbb{P}(z_d = 0) = 1 - \kappa$ and $\mathbb{P}(z_d = \pm \bar{z}) = \kappa/2$.
- b_0 and $\bar{b}$ are the initial and maximum stake, $0 \leq b_0 \leq \bar{b}$, with $b_0 < \tau$ maintained: a pre-existing crossing is outside the core.
- $\mathcal{J}$ is a finite ordered menu of complete contingent plans, least to most aggressive, $|\mathcal{J}| = J < \infty$; $a_j \in \{0, 1\}$ is the engagement attached to plan j , with $a_j = 1$ for Voice plans and 0 for Exit and Hold.
- $B_j(s, d) \in [0, \bar{b}]$ is the cumulative pooled stake at business day d , with $B_j(s, -1) = b_0$; for Voice plans $\partial_d B_j \geq 0$ and $\partial_s B_j \geq 0$, Hold is constant and Exit weakly decreasing. For *every* plan and every d , $s \mapsto B_j(s, d)$ is Borel — automatic for Voice and Hold, a genuine requirement for Exit, and needed because pooled pricing integrates over every type including Exit types.
- $b_j^*(s) = B_j(s, H)$ is the terminal target stake; H is the finite control-decision horizon in business days.
- $c_j(s; \tau) = \inf\{d : B_j(s, d) \geq \tau\}$ is the threshold-crossing date ($+\infty$ if it never happens); $f_j = c_j + T$ is the legal filing date; the flag lands iff $f_j \leq H$, equivalently iff $B_j(s, H - T) \geq \tau$.
- $D_j(s; \tau, T) = \mathbf{1}\{a_j = 1, c_j < \infty, f_j \leq H\}$; $D = 1 \Rightarrow a = 1$. $B_j^F = B_j(s, f_j)$ is the stake at filing and $Q_j^F = b_j^*(s) - B_j^F \geq 0$ the flagged-round order for a Voice plan.
- Γ is a finite ordered coarsening from stake increment to pooled order mark, $q_{jd}(s) = \Gamma(B_j(s, d) - B_j(s, d - 1))$, and observed pooled order flow is $X_d = q_{jd} + z_d$. (Γ , not ψ : ψ is tender-game pivotality and χ is the frozen manuscript's cost parameter.)

Information, prices, control outcome

$\mathcal{H}_d^P = (X_0, \dots, X_d; \text{flag landed by } d)$ is the pooled public history, finite. $F = (B^F, a = 1)$ is the truthful filing message and $\mathbf{S}_F = (B^F, Q^F, a = 1)$ is F augmented by the flagged-round order. The control-node information set is

$$\mathcal{I}_H = \mathcal{H}_H^P \text{ on the pooled cell } \{D = 0\}, \quad \mathcal{I}_H = \mathbf{S}_F \text{ on the flagged cell } \{D = 1\},$$

the bidder's own ξ being private. The engagement posterior is $\pi(\mathcal{I}) = \mathbb{P}(a = 1 \mid \mathcal{I})$, equal to 1 on the flagged cell; entry probability is $p(\mathcal{I}) = 1 - \Phi((P + K + m_0 + \pi\Delta_m - \bar{S})/\sigma_\xi) \in (0, 1)$; and with entry indicator $\mathbf{B}$ the terminal shareholder payoff is

$$Y = (1 - \mathbf{B})(v + a\Delta_V) + \mathbf{B}(P + m_0 + a\Delta_m).$$

Prices are competitive, $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$, with the convention $P_{-1}^P := \mathbb{E}[Y]$ for the pre-trading pooled price (needed whenever $c = 0$, which $T = H$ forces on every flagged history). The genuine fixed point sits at control nodes: at an earlier pooled date $d < H$ the price is a tower expectation of already-solved control-node values, with no self-reference. $P_{\text{ND}}(\mathcal{H}_{f-}^P) = P_{f-}^P$ is the *not-yet-disclosed* price at the same realised order flow — not a never-disclosed counterfactual, under which reading the run-up/jump identity acquires a residual term. Then

$$R_d = P_d^P - P_{c-}^P, \quad R = P_{f-}^P - P_{c-}^P, \quad J = P^F - P_{\text{ND}}, \quad P^F - P_{c-}^P = R + J \text{ exactly.}$$

J is *not* claimed κ -invariant.

The blockholder's objective

$$U_j(s) = \mathbb{E}[b_j^*(s)Y - \mathcal{C}_j^{\text{trade}} - a_j C_j(s) \mid s, j],$$

the expected terminal value of the position the plan builds, net of what it costs to build and to engage: $\mathcal{C}_j^{\text{trade}}$ is the execution outlay (increments valued at the pooled prices P_d^P up to the

plan's last pooled date, plus $Q_j^F(s)P^F$ when $D_j = 1$) and $C_j(s) \geq 0$ the engagement cost. Only two properties of U_j are ever used: *plan-locality* — U_j depends on j only through the executed stake path, the prices paid on it, the terminal stake, the engagement flag and the cost — and *integrability*, $\mathbb{E}[\max_j |U_j|] < \infty$.

Premium and sensitivity objects

$$\begin{aligned} h(\mathcal{I}) &= \pi(\mathcal{I})p(\mathcal{I}), \quad h \geq 0, \quad h(0) = 0; & \Delta^{\text{act}} &= \Delta_m \mathbb{E}[h(\mathcal{I}_H)] \geq 0; \\ M_F &= \Delta_m \mathbb{E}[h \mid D = 1], \quad M_P = \Delta_m \mathbb{E}[h \mid D = 0]; & \Omega &= \mathbb{P}(D = 1) = \mathbb{P}(a = 1) \omega_a; \\ \mathcal{S} &= |\partial_\kappa \Delta^{\text{act}}|, \quad \mathcal{S}_P = |\partial_\kappa M_P|; & C_h(\bar{\pi}) &= h(0) - 2h(\bar{\pi}/2) + h(\bar{\pi}). \end{aligned}$$

$\omega_a = \mathbb{P}(D = 1 \mid a = 1)$ is the disclosed share of engagements, the calibration target; Ω is the unconditional flagged weight and is the frozen manuscript's ω_P . $\bar{\pi}$ is the **upper support point** of the pooled engagement posterior in the $A(\tau)$ representation — *not* the pooled engagement share. The share is the mean $\mathbb{E}[\Pi_\kappa]$, which under $A(\tau)$ is κ -invariant (a mean-preserving spread) and is strictly below $\bar{\pi}$ in any non-degenerate case, equalling $\bar{\pi}/2$ only under level symmetry $A_0 = A_1$. Reading $\bar{\pi}$ as the mean forces a point mass at $\bar{\pi}$ with $A'_\kappa = 0$ and zero interior motion — degenerate, and excluded. A'_κ is the common derivative of the $A(\tau)$ weights ($A'_0 = A'_1 = A'_\kappa$, $A'_{1/2} = -2A'_\kappa$). Weight-effect ratios are W_τ, W_T (e.g. $W_T = (1 - \Omega(\tau, 5))/(1 - \Omega(\tau, 10))$) and composition-effect ratios are C_τ, C_T (e.g. $C_T = \mathcal{S}_P(\tau, 5)/\mathcal{S}_P(\tau, 10)$), unsigned.

For the general-equilibrium certificate: $k = (k_1 \leq \dots \leq k_{J-1})$ is the cutoff vector on the compact ordered polytope Θ with parameter vector ϑ ; $\mathcal{T}(k; \vartheta)$ the outer best-response map; $L_{\mathcal{R}} = \sup_{\mathcal{R}} \|D_k \mathcal{T}\|$ the contraction bound; $r_\tau = -\tau$, $r_T = -T$ the strictness coordinates (higher r = tighter); $g_r^{PE} = -\text{sgn}(d\Delta^{\text{act}}/d\kappa) \partial_{\kappa r} \Delta^{\text{act}}$ the direct fixed-policy attenuation margin; $\bar{k}_x = |\partial_x \mathcal{T}|/(1 - L_{\mathcal{R}})$ and $\bar{k}_{\kappa r}$ the inversion-free derivative bounds;

$$\mathcal{B}_r^{GE} = |\Delta_{\kappa k}| \bar{k}_r + (|\Delta_{kr}| + |\Delta_{kk}| \bar{k}_r) \bar{k}_\kappa + |\Delta_k| \bar{k}_{\kappa r}, \quad \eta_r = g_r^{PE} - \mathcal{B}_r^{GE}$$

the GE remainder bound and the slack on a certified region $\mathcal{R}_r$.

3 Timing: the two rounds

1. Nature draws v and the blockholder's signal s ; the blockholder picks one complete contingent plan j from the finite ordered menu $\mathcal{J}$.
2. **Round 1 — pooled trading.** The plan's stake path executes over business days $d = 0, \dots, H$. Market makers see pooled order flow and set $P_d^P = \mathbb{E}[Y \mid \mathcal{H}_d^P]$. There is **no within-window re-optimisation**, and hence no feedback from realised order flow or prices into the path: $B_j(s, d)$, $q_{jd}(s)$ and Q_j^F are functions of (j, s, d) and (j, s, τ, T) alone. This is a numbered hypothesis, not a modelling aside: two steps of L2 fail without it.
3. **Disclosure node.** The flag lands iff $D = 1$, i.e. iff the plan engages, crosses τ at some date $c < \infty$, and $c + T \leq H$. The filing reveals $F = (B^F, a = 1)$. **The flag terminates the pooled round.** Pooled trading stops when the filing lands; the flagged round follows it and the bidder acts after that. Without this reading $Q^F = b_j^*(s) - B_j^F(s)$ is not the blockholder's whole residual position, and the flagged-round step of P1 fails.
4. **Round 2 — flagged trading, then the bidder.** If $D = 1$ the blockholder submits Q^F , the market prices $P^F = P(F, Q^F)$, and then the bidder decides. If $D = 0$ there is no flagged round and the bidder acts on the pooled history.

Sequence: pooled round $\rightarrow$ flag or no flag $\rightarrow$ flagged round if applicable $\rightarrow$ bidder decision.

4 The partition

The disclosure indicator $D = \mathbf{1}\{a = 1, c(\tau) + T \leq H\}$ maps every control-node history into exactly one of two cells, $\mathcal{C}_F$ (flagged) and $\mathcal{C}_P$ (pooled), exclusive and exhaustive by construction. The cells are keyed by the disclosure rule and by nothing else:

- On $\mathcal{C}_F$ the control-node information set is the flagged tuple $\mathbf{S}_F = (B^F, Q^F, a=1)$: the market knows engagement happened, $\pi = 1$, and the pooled residual is (under A7') pure noise conditional on $\mathbf{S}_F$.
- On $\mathcal{C}_P$ the control-node information set is the pooled public history $\mathcal{H}_H^P$: the market has an inference problem, and the posterior $\pi(\mathcal{H}_H^P)$ is where κ enters.
- The clock equivalence: for every Voice plan, $f_j \leq H \iff B_j(s, H - T) \geq \tau$. A tighter threshold (lower τ) or a tighter window (lower T) moves histories from $\mathcal{C}_P$ into $\mathcal{C}_F$ and nowhere else.
- Each flagged history yields B^F, R_d, R, J , with $P^F - P_{c-}^P = R + J$ exactly.
- $\Omega = \mathbb{P}(D = 1)$ is the cell weight; A8 asks for $0 < \Omega < 1$, which is required only for positive cell mass, never for the structural partition itself.

5 Equilibrium notion

Cutoff perfect Bayesian equilibrium: (i) a weakly ordered cutoff vector $k = (k_1 \leq \dots \leq k_{J-1})$ mapping s into a plan; (ii) sequentially optimal pooled and flagged components; (iii) Bayes-consistent beliefs on path; (iv) competitive pooled and flagged prices at their fixed points; (v) the bidder-entry rule; (vi) off-path beliefs as limits of full-support perturbations. Weak inequalities permit collapsed action regions, Hold included. Existence is Brouwer on the compact ordered polytope Θ for the outer map $\mathcal{T}(k; \vartheta)$, i.e. $k = \mathcal{T}(k; \vartheta)$. **Uniqueness is not claimed.**

6 Standing hypotheses

- A1 Independent primitives.** v, ε, ξ and all z_d mutually independent; all variances strictly positive.
- A2' Finite model, amended boundedness.** Finiteness is unchanged: the plan menu $\mathcal{J}$, the image of Γ , the noise support $\{-\bar{z}, 0, +\bar{z}\}$ and the horizon H are all finite. The old flat boundedness clause was *false* and is replaced by: prices and payoffs are locally bounded in (s, ϑ) on the maintained parameter set, and $\mathbb{E}[\max_{j \in \mathcal{J}} |U_j|] < \infty$ for every $k \in \Theta$. Flat global boundedness is inconsistent with the rest of the model — v is Gaussian and the flagged region is unbounded in s under A7', so Y , prices and U_j are unbounded. Integrability is all any proof consumes.
- A3 Ordered plans, single crossing.** At every belief/price system, adjacent-plan payoff differences cross zero at most once in s , and the preferred plan is weakly increasing in s .
- A4 Legal-clock discipline.** c is the first date the path reaches τ ; the filing lands exactly at $c + T$; filings truthfully reveal stake and purpose; only Voice plans cross in the core.
- A5 Inner pricing regularity, mostly demoted to a theorem.** Each public-history pricing map has a unique fixed point, continuous in beliefs, cutoffs and parameters. *Under $m_0 \geq 0$ — a card restriction — existence, uniqueness and continuity of the inner fixed point are theorems, not assumptions:* with $\bar{y}(\mathcal{I}) = \mathbb{E}_\mu[v] + \pi\Delta_V$ and $\bar{m}(\mathcal{I}) = m_0 + \pi\Delta_m$, the pricing map is $P \mapsto \bar{y} + \bar{m}p(P)/(1 - p(P))$, strictly decreasing in P wherever $\bar{m} \geq 0$, so it crosses the identity exactly once. A5 is retained only as its continuity clause: the pricing family is continuous in the cutoff vector and the parameters and measurable in the flagged tuple. Because the flagged information sets are continuum-indexed, “unique fixed point” must be read as a measurably selected *family*, not a finite list.

A6 Compact outer self-map. All best-response cutoffs lie in a common compact ordered polytope Θ ; $\mathcal{T}$ is continuous and maps Θ into itself.

A7 / A7' Filing sufficiency. On flagged histories $(B^F, Q^F, a = 1)$ identifies the informed component of the selected plan; conditional on it the pooled order-flow residual is pure noise, independent of (v, s, ξ) . *L2's proof uses the injective form and the weak wording is not sufficient* — it permits two (j, s) pairs with different pooled paths, which is L2's first failure case. The injective form is A7': on the flagged set the composed terminal target $s \mapsto b_{j(s)}^*(s)$ is strictly increasing *for every cutoff vector* $k \in \Theta$; for a menu this amounts to each b_j^* strictly increasing with no backtracking across any admissible plan switch. Strictness of B^F is neither necessary (it fails at crossing-date jumps on the pro-rata menu) nor sufficient (multi-Voice backtracking). Injectivity forces B^F continuum-valued; injective plus measurable already gives a measurable inverse. *Satisfiability is resolved:* A7' plus a fixed cutoff policy plus $\Omega > 0$ deliver the on-path injective form with an explicit inverse, and a satisfying menu exists (the pro-rata single-Voice menu with terminal target strictly increasing on all of $\mathbb{R}$, which also satisfies the joint (j, s) form). Failure boundary: a binding stake cap, quantized stakes, a composed target repeating values across Voice-plan switches, $\Omega = 0$, and policy-dependence when the condition is stated at only one equilibrium's cutoffs. A7' menus are fully separating on the flagged set, so the burden moves to incentive compatibility rather than away.

A8 Interior crossing. $0 < \Omega(\kappa, \tau, T) < 1$.

A(τ) Threshold chord restriction. The pooled posterior law has the symmetric ternary representation

$$\mathbb{E}[h] = A_0(\kappa)h(0) + A_{1/2}(\kappa)h(\bar{\pi}/2) + A_1(\kappa)h(\bar{\pi}), \quad A'_0 = A'_1 = A'_\kappa, \quad A'_{1/2} = -2A'_\kappa,$$

with maintained orientation $C_h(\bar{\pi}) \leq 0$ and $|C_h|$ weakly increasing in $\bar{\pi}$. Two clauses:

- (τ -i) *The kernel depends on the information set only through the engagement posterior:* $h(\mathcal{I}) = h(\pi(\mathcal{I}))$, so $h(0)$, $h(\bar{\pi}/2)$, $h(\bar{\pi})$ are well defined and κ -free. This is a restriction, not a reading: $h = \pi p$ and p depends on the price as well as on π , so in the model $h = \pi p(\hat{v}, \pi)$ is a function of *two* scalars. The clause says the standalone-value channel and the engagement channel do not co-move inside the pooled cell in a way that moves h at a fixed posterior.
- (τ -ii) *The support and $\bar{\pi}$ are κ -free; only the weights move.* Without the second half, L3's conclusion is **false**: the derivative gains a term first order in $\bar{\pi}$ and the vanishing fails.

Where the bite is: given a κ -invariant three-point support, the derivative restrictions are equivalent to κ -invariance of the pooled block's total mass and of its unnormalised engagement moment, both of which the model delivers at fixed policies. A(τ)'s entire remaining content is the support condition. A one-round ternary-noise market with informed mark $2\bar{z}$ and pre-order engagement share $\frac{1}{2}$ satisfies it; the frozen manuscript's own no-disclosure structure (informed mark $\bar{z}$) does **not** — its pooled law has four atoms, two of which move with κ .

Whether the two-round pooled cell of the timing above satisfies the support condition is OPEN. Every A(τ)-conditional result — and therefore L4 leg 3 and T1 Part B — inherits that conditionality.

A(br) Chord-sensitivity bridge. Consumed by L4 leg 3 and T1 Part B, and by nothing else.

For two compared thresholds $\tau' < \tau$ at fixed policies and a common κ :

- (br-i) *Representation at both policies.* A(τ)'s symmetric ternary representation holds for the pooled class under τ and under τ' , with endpoints $\bar{\pi}(\tau)$, $\bar{\pi}(\tau')$ and coefficients $A'_\kappa(\tau)$, $A'_\kappa(\tau')$.
- (br-ii) *κ -localisation.* At fixed policies all κ -dependence of M_P sits in the A(τ) weights, so $\partial_\kappa M_P = \Delta_m A'_\kappa C_h(\bar{\pi})$ exactly, with no composition-through- κ remainder. Written against the honest reading $h = \pi p(\hat{v}, \pi)$, this repairs an ambiguity rather than adding a fourth independent restriction; the trailing “hence” is derivable, not assumed.

- (br-iii) *Coefficient stability across the threshold margin.* $|A'_\kappa(\tau')| \leq |A'_\kappa(\tau)|$; weakest sufficient form is equality — reclassification changes *which* histories are pooled, not the κ -responsiveness of the pooled weights. *This is the clause with the least justification behind it; it is the one to attack first.*
- (br-iv) *Endpoint linkage.* $\bar{\pi}$ is $A(\tau)$'s chord endpoint — the upper support point — and is a weakly increasing function of the pooled prior engagement share $\bar{\pi}_{\text{pr}} = \mathbb{P}(a = 1 \mid D = 0)$, the same function at τ and τ' . The identity branch $\bar{\pi} = \bar{\pi}_{\text{pr}}$ is excluded as degenerate.
- (br-v) *Comparability of the chord functional across thresholds.* $C_h(\cdot)$ — and the kernel h it is built from — are the same functions of the posterior at both compared thresholds. Without it, leg 3 compares $|C_h|$ across two different functionals and the comparison is meaningless. Required independently by three agents, and confirmed not implied by (br-i)–(br-iv).

Sharpening on file, recorded not assumed: $\bar{\pi} = \bar{\pi}_{\text{pr}}/\rho$ with $\rho := \frac{1}{2}A_{1/2} + A_1$ provably κ -free, so (br-iv) is equivalent to $\rho(\tau')/\rho(\tau) \geq \bar{\pi}_{\text{pr}}(\tau')/\bar{\pi}_{\text{pr}}(\tau)$. Under the level-symmetric reading $\rho = \frac{1}{2}$ and $\bar{\pi} = 2\bar{\pi}_{\text{pr}}$, forcing $\bar{\pi}_{\text{pr}} \leq 1/2$ — an inherited restriction on $A(\tau)$'s domain that is not resolved.

AGE GE differentiability and contraction. On a candidate region $\mathcal{R}$ the outer map is twice continuously differentiable, $L_{\mathcal{R}} < 1$, and the sign of the equilibrium liquidity derivative is constant on $\mathcal{R}$.

7 The result ledger

Seven of eight results carry two-pass evidence — an adversarial proof-read PASS *and* an independent statements-only re-derivation PASS, by different agents — and are labelled PROVED with their conditionality written into the label. C1 is CONJECTURE. Labels are transcribed from LABEL_LEDGER.md; each statement is the amended one, with its hypothesis set named in full. Nothing was weakened silently.

ID	Statement, with its full hypothesis set	Label
D1	Under A1, A2', A4, A5 , the primitive/plan table restrictions, the cutoff selection map, and the pricing conventions ($P_{-1}^P = \mathbb{E}[Y]$; $P_{\text{ND}} = P_{f-}^P$), plus two hypotheses this regeneration wrote onto the card — Borel regularity of $s \mapsto B_j(s, d)$ for every plan including Exit (needed only for part (c), since pooled pricing integrates over all types) and a content for $\mathcal{I}_H$ — the indicator $D = \mathbf{1}\{a = 1, c(\tau) + T \leq H\}$ is measurable and maps every control-node history into exactly one cell; for every Voice plan $f_j \leq H \iff B_j(s, H - T) \geq \tau$; and each flagged history yields B^F, R_d, R, J with $P^F - P_{c-}^P = R + J$.	PROVED
L1	Under D1 , the premium definitions, A5 (which pins <i>the</i> version of $\mathbb{E}[Y \mid \mathcal{I}]$), A2' with Δ_m finite , and A1 (one probability space): whenever $0 < \Omega < 1$, $\Delta^{\text{act}} = \Omega M_F + (1 - \Omega)M_P$; at $\Omega = 1$ it degenerates to $\Delta^{\text{act}} = M_F$ and at $\Omega = 0$ to $\Delta^{\text{act}} = M_P$, the null-cell average being <i>undefined rather than imputed</i> — proved as a non-identification statement, not asserted.	PROVED

ID	Statement, with its full hypothesis set	Label
L2	At fixed cutoff and execution policies, under A1; A2' with the primitive/plan table restrictions; A4; A5; A7' in its on-path injective form, consumed almost surely on the flagged set; D1; the no-feedback timing; and $\Omega > 0$ — together with an explicit bidder-entry rule carried as bookkeeping: $(B^F, Q^F, a=1)$ makes the pre-filing pooled history conditionally independent of (v, s, ξ) on the flagged set, so the flagged posterior, price, entry probability and M_F are invariant to κ . (“Almost surely” is the only coherent reading once B^F is continuum-valued, since then no individual flagged tuple has positive probability.)	PROVED
L3	Under $A(\tau)$ — including $(\tau\text{-i})$ kernel-through-posterior and $(\tau\text{-ii})$ κ -free support <i>and</i> κ -free $\bar{\pi}$ — plus $h(0) = 0$; κ -free pooled mass and engagement moment at fixed policies; D1 by statement; minimal regularity (h continuous on $[0, \bar{\pi}]$, twice differentiable on the open interval — Darboux does the rest, no continuity of h''); for the small- $\bar{\pi}$ corollary only, a second-order Peano expansion of h at $0+$ and one and the same kernel along the shrinking family; and, for the seam where L4 consumes L3, $ A'_\kappa $ bounded <i>uniformly in $\bar{\pi}$</i> . Then $\partial_\kappa \mathbb{E}_\kappa[h] = A'_\kappa C_h(\bar{\pi})$ exactly; $C_h(\bar{\pi}) = \frac{1}{4}\bar{\pi}^2 h''(\zeta)$ for some $\zeta \in (0, \bar{\pi})$ — an identity, not an approximation; and $C_h = \frac{1}{4}h''(0)\bar{\pi}^2 + o(\bar{\pi}^2)$, so the interior motion vanishes at rate $\bar{\pi}^2$. An “if”, never an “iff” ($A'_\kappa = 0$ also kills the motion). Whether the two-round pooled cell satisfies $A(\tau)$ ’s support condition is OPEN .	PROVED under $A(\tau)$
L4	At fixed policies, for $b_0 < \tau' < \tau$ at a common window T and common κ , with $\Omega(\tau', T) < 1$: (leg 1, unconditional) $\mathcal{C}_F(\tau, T) \subseteq \mathcal{C}_F(\tau', T)$ with every newly flagged history generated by a Voice plan, hence $\Omega(\tau', T) \geq \Omega(\tau, T)$; (leg 2, unconditional) the pooled engagement <i>share</i> falls, $\bar{\pi}_{\text{pr}}(\tau') \leq \bar{\pi}_{\text{pr}}(\tau)$, with an exact identity for the gap; (leg 3, under $A(\text{br})$) $\mathcal{S}_P(\tau', T) \leq \mathcal{S}_P(\tau, T)$, with equality whenever $C_h(\bar{\pi}(\tau)) = 0$. Legs 1–2 need only D1’s clock equivalence, the no-feedback timing, fixed policies, $b_0 < \tau' < \tau$ imposed at both thresholds, A1, A4, $D = 1 \Rightarrow a = 1$, and $\Omega(\tau') < 1$ — the “nestedness” clauses are <i>conclusions, not hypotheses</i> . Leg 3 additionally needs L3 by statement, $A(\tau)$’s maintained magnitude monotonicity of C_h (the sign half $C_h \leq 0$ is never used at this leg) and $A(\text{br})$ (br-i)–(br-v) .	PROVED (legs 1–2 outright; leg 3 under $A(\text{br})$)

ID	Statement, with its full hypothesis set	Label
P1	Under A1, A2', A3, A4, A6, A7' (on-path injective), D1 by statement, the no-feedback timing read with the flag-terminates-the-pooled-round clause, the definitional round-2 action-set hypothesis (the flagged-round action set <i>is</i> the plan-generated set $\{Q_{j'}^F(s)\}$ over menu elements agreeing with j on everything already played — <i>not</i> a closure condition, which is jointly unsatisfiable with finiteness by cardinality), $m_0 \geq 0$, and the blockholder-objective definition U_j : a cutoff PBE over complete contingent plans exists — $k^* \in \Theta$ with $k^* = \mathcal{T}(k^*; \vartheta)$, prices at their inner fixed points, Bayes-consistent on-path beliefs, off-path beliefs as limits of full-support perturbations over <i>plans</i> , the entry rule, and a sequentially optimal flagged component. A5 is not assumed : its existence and uniqueness content is derived from $m_0 \geq 0$. At any such equilibrium at which A8 holds , both cells carry strictly positive probability and are on path; for A8's restatement as a single signal threshold add H-ord (Voice stake monotonicity across plans) and the upper-set engagement-flag hypothesis. A8 is a condition <i>on the fixed point exhibited</i> : existence of an equilibrium <i>at which</i> A8 holds is not claimed. Uniqueness is not claimed.	PROVED
T1	At fixed plan and cutoff policies, with $0 < \Omega < 1$ and $\mathcal{S}_P > 0$: (A) $\mathcal{S} = (1 - \Omega)\mathcal{S}_P$ exactly, and the same factorisation holds for the total-variation aggregate of Δ^{act} over any κ -grid with no differentiability required; (B) threshold tightening attenuates — $\mathcal{S}(\tau')/\mathcal{S}(\tau) = W_\tau C_\tau \leq 1$ because <i>both</i> ratios lie in $[0, 1]$, no dominance condition needed; (C) window tightening attenuates iff $W_T C_T \leq 1$, where $W_T \leq 1$ is <i>proved</i> (from D1's clock equivalence and the monotone Voice stake path) and C_T is <i>unsigned</i> — “equivalently $\partial_{r_T} \mathcal{S}_P / \mathcal{S}_P \leq \Omega_{r_T} / (1 - \Omega)$ ” holds on average along the tightening path (integrated over $[-T, -T']$), exactly in the infinitesimal limit, and is false read pointwise . Hypotheses: fixed policies; A8 at each compared policy; $\mathcal{S}_P > 0$; L1; L2 (its own hypotheses travelling); D1; PE-Ω ($\partial_\kappa \Omega = 0$ at fixed policies — derivable, not assumed, and it fails in GE, which is C1's term); κ-differentiability of M_P (no card hypothesis supplies this — carried in-proof); $A(\tau)$ at both compared policies with the $\bar{\pi}$ ruling; L3; $A(\text{br})$ (br-i)–(br-v) at the threshold pair; L4; the no-feedback timing; a smooth window interpolation for the local form; threshold-side smoothness (confirmed non-load-bearing). No unconditional window sign is claimed.	PROVED at fixed policies
C1	On a named region where $L_{\mathcal{R}} < 1$ and $g_r^{PE} > \mathcal{B}_r^{GE}$, the fixed-policy attenuation sign survives in equilibrium. The certificate theorem is two-pass complete — an adversarial proof-read PASS (0 FAIL, 13 repairs, 7 observations) and an independent statements-only re-derivation PROVED-WITH-CHANGES (a norm convention and two-sided openness of $\mathcal{R}_r$ added; at $J - 1 \geq 2$ the bare $ \cdot $ in $\mathcal{B}_r^{GE}$ is not well defined and a mismatched reading makes part (B) false) — and the executed 80-node certification returns a nonempty region: 18 nodes certified , slack η_r from 0.0595 (min) through 0.3467 (median) to 1.7227 (max), $L_{\mathcal{R}} \leq 0.5008$ at every node. The ledger decision is pending , so the label does not move.	CONJECTURE

The remaining aspiration, not a claim: C1 PROVED on a named nonempty region, NUMERICAL off-region, dropped if the region is empty.

Labels

PROVED — a complete proof, independently re-derived and proof-read. **NUMERICAL** — verified on a grid by an executed, committed check script with committed output. **ESTIMATED** — an empirical estimate with a standard error and a stated design. **CONJECTURE** — everything else, including anything whose proof is deferred. Region-certified is not a fifth label: it is PROVED with the region named in the hypothesis. Labels are never weakened by editing; only an executed check or an independent re-derivation may move one.

7.1 Three known facts on file

These are facts the record carries, not claims the model makes. They are the reason the labels above are written with their conditions attached.

1. The window analogue of L4 leg 3 is refuted at the O-1 calibration. Since $W_T \leq 1$ is proved, the O-1 evaluations

$$W_T C_T = 1.06397 / 1.18373 / 1.13631 \quad \text{at} \quad \Omega = 0.037252 / 0.128950 / 0.285804$$

force $C_T \geq W_T C_T > 1$: window tightening *raises* pooled sensitivity at those nodes, the opposite of leg 3’s threshold-side direction. At $\Omega = 0.50$ the criterion holds ($W_T C_T = 0.37798$), with the boundary located by bisection at $\Omega^* \approx 0.3428$ ($k_D^* = 1.28618$). Both branches occur inside the maintained parameter set, which is the sharpest possible content of “no unconditional window sign”. Provenance: this is the referee’s O-1 experiment run on the *repo* model (the frozen manuscript’s static structure, fixed cutoffs, partial equilibrium), re-executed in `quality_reports/fixes/t1_o1_rerun_check.py` with every committed number reproducing to the last printed digit; it is not the two-round build.

2. The implemented two-round pooled cell FAILS $A(\tau)$ at baseline. Two executed checks measure the same gap, and both report it rather than smoothing it.

- `t2_l2_check.json`, check `l2_placebo_M_P_sign_A_tau`: **FAIL**. The enumerated pooled M_P is *hump-shaped* in κ — 10 of 18 increments positive, one sign change, peak near $\kappa = 0.55$ — so $\partial_\kappa M_P \leq 0$, the orientation $A(\tau)$ ’s maintained $C_h \leq 0$ would give, does not hold globally. The JSON’s own words: a failed hypothesis about $A(\tau)$ ’s orientation on the enumerated two-round pooled law, not a statement about L2.
- `t2_t1_check.json`, check `t1_block3_chord_magnitude`: **FAIL**. The residual $|\mathcal{S}_P - \Delta_m |A'_\kappa| |C_h(\bar{\pi})|$ is 0.006279 (0.6279 premium percentage points) against a predicted 10^{-10} — the gap between the *enumerated* two-round pooled sensitivity and $A(\tau)$ ’s three-atom closed form. (The companion clause of the same check passed: the ratio $|C_h|/\bar{\pi}^2$ is constant to 0.48% between the two smallest $\bar{\pi}$ nodes, well inside the 5% criterion.)

This is exactly the open question flagged under $A(\tau)$ above, now with a number attached: the representation’s support condition is not satisfied by the implemented pooled law at this calibration. It is a failed hypothesis, not a wiring error — the same gap is measured analytically on the frozen manuscript’s four-atom law by `t2_l3_check.py` block 5b.

3. L4’s sign predictions hold numerically with zero violations. `t2_l4_check.json` reports 10 checks, 0 failures. Across all 16 tightening steps and both windows: Ω rises at every step (0 violations, minimum increment +0.02263); $\bar{\pi}_{\text{pr}}$ falls at every step (0 violations, largest increment -0.018766); and $\mathcal{S}_P$ falls at every step (0 violations, largest increment -9.47×10^{-5}) — and, as the JSON notes, a single violation would be a failed hypothesis rather than sampling error, because nothing in that computation is sampled. Ω and $\bar{\pi}_{\text{pr}}$ are flat in κ to exactly zero across 71 grid points, which is what the no-feedback timing predicts.

What the pair of facts means. The chord mechanism is **hypothesis-bound**: it needs $A(\tau)$, and $A(\tau)$ is not satisfied by the implemented two-round pooled cell at baseline. The threshold phenomenon is **not**: the reclassification legs of L4 are unconditional, and they hold numerically without exception.

8 What this note does not claim

A global window-margin attenuation sign; κ -invariance of J ; equilibrium uniqueness; a nonempty GE-certified region *as a theorem* (the 18 certified nodes are an executed record, and C1's ledger decision is pending); endogenous filing before the deadline; noisy or partially revealing flagged-round trading; continuous-time execution; welfare or optimal rule design; that the frozen manuscript's hump result survives; that the prior calibration ($\Omega \approx 0.037$) is economically meaningful; any empirical value for ω_a .

Three items are explicitly open:

1. **Whether the two-round pooled cell satisfies $A(\tau)$ — OPEN.** L3 proves the representation's entire remaining bite is the support condition, exhibits a one-round market that satisfies it and the frozen manuscript's own no-disclosure structure that does not, and names the weakest sufficient conditions. L3, L4 leg 3 and T1 Part B are all conditional on $A(\tau)$, so this is the largest single conditionality the ledger carries — and the executed checks of fact 2 above show it failing on the implemented law at baseline.
2. **Whether an equilibrium in which the blockholder chooses the fully separating plan exists on a given calibration — OPEN, and P1-adjacent.** A7' menus are fully separating on the flagged set, so the burden did not disappear when A7 satisfiability was resolved; it *moved* to incentive compatibility, which P1 does not settle.
3. **The window-margin analogue of L4 leg 3 is refuted at the O-1 calibration** (fact 1 above).

9 What v4 takes from draft_v2, what it simplifies, what it drops

The frozen manuscript `draft_v2.tex` is a record the supervisor has seen; it is not edited and it is not cross-referenced here. The table below describes its objects rather than citing them.

draft_v2 object	Verdict	What carries over, what changes
Four-action structure: Exit, Hold, Quiet Voice, Public Voice, with weakly ordered cutoffs (k_1, k_0, k_D) ; and the dominance lemma making the four-action menu a result rather than an assumption (engaged exit and silent buy are never chosen).	REUSE	Becomes the <i>special case</i> of the v4 ordered plan menu: $\mathcal{J}$ finite and ordered with $J \geq 3$, cutoff vector $k = (k_1, \dots, k_{J-1})$, which maps to (k_1, k_0, k_D) exactly when the menu is the four named actions. What changes: menu elements are now <i>complete contingent plans</i> over the calendar $d = 0, \dots, H$, not single one-shot (q, a) pairs, and A3's single crossing is stated for adjacent plans. The dominance argument survives as the reason the menu is ordered and finite.

draft_v2 object	Verdict	What carries over, what changes
Brouwer existence architecture: boundedness of best responses, the self-map and order-preservation lemma, and existence of a monotone equilibrium by Brouwer on a nonempty compact convex Θ .	REUSE	Reused whole as P1's architecture: Θ compact, ordered, convex; $\mathcal{T}$ continuous $\Theta \rightarrow \Theta$; $k^* = \mathcal{T}(k^*; \vartheta)$; collapse faces permitted (the manuscript's own baseline collapses Hold). What changes: the fixed point ranges over plans; off-path beliefs are limits of full-support perturbations <i>over plans</i> ; and the inner pricing fixed point is no longer an assumption (next row).
The (C*) chord condition $\mathcal{C}(\bar{\pi}) = h(0) - 2h(\bar{\pi}/2) + h(\bar{\pi})$, and the lemma signing the posterior/pricing channel under it.	REUSE	Is v4's $C_h(\bar{\pi})$, same expression, same maintained orientation $C_h \leq 0$ with $ C_h $ weakly increasing in $\bar{\pi}$. What changes: in draft_v2 it selects hump versus trough in κ ; in v4 it is the magnitude that drives pooled sensitivity through L3's exact identity $\partial_\kappa \mathbb{E}_\kappa[h] = A'_\kappa C_h(\bar{\pi})$, and the $A(\tau)$ representation that licenses that identity is stated as a restriction whose support condition is open.
The tender-game microfoundation of the premium wedge (the disagreement-node continuation deriving the appropriability coefficient $\lambda = 1 - q(1 - \gamma)\psi$ and the resulting wedge).	REUSE	Kept as the <i>input</i> that supplies (m_0, m_1) . v4 takes m_0, m_1 as primitives with $m_1 > m_0$ and $m_0 \geq 0$ and does not re-derive them; the tender game is where those numbers come from and why $\Delta_m > 0$ is not free. Its symbols are reserved and unavailable: bare λ is the appropriability coefficient, ψ is pivotality.
The GE dominance-check pattern: contraction modulus L along the path, an inversion-free Neumann bound $\bar{B}$, pointwise dominance off a ball plus a ball-integral condition, a certified interval, and a counterexample showing the global claim false.	REUSE	Is C1's precedent. $\mathcal{B}_r^{GE}$ is the <i>cross-derivative</i> analogue of $\bar{B}$; $\eta_r = g_r^{PE} - \mathcal{B}_r^{GE}$ is the slack; certification is node by node on a named region, and an empty region is a reportable outcome that would drop C1. What changes: the object bounded is a cross-derivative in (κ, r) rather than a single-argument profile, and the strictness coordinate $r_T = -T$ is not differentiable because the window is an integer.
Stake-triggered disclosure as the market's partition, with the flag a function of the order ($D = \mathbf{1}\{q = +1\}$: a buy cannot be concealed).	REUSE	The idea that the rule <i>is</i> the partition is the whole v4 position. What changes is the trigger: $D = \mathbf{1}\{a = 1, c(\tau) + T \leq H\}$, a stake threshold plus a filing window on a business-day calendar, with D1's clock equivalence $f \leq H \iff B(s, H - T) \geq \tau$. The window T becomes a genuine primitive instead of a flag that is either on or off.

draft_v2 object	Verdict	What carries over, what changes
Assumption (A2)'s flat boundedness of prices and payoffs.	SIMPLIFY	Replaced by A2': finiteness clauses kept verbatim; boundedness weakened to local boundedness plus integrability $\mathbb{E}[\max_j U_j] < \infty$. The flat bound was <i>false</i> in this model (<i>v</i> Gaussian, flagged region unbounded in <i>s</i>), and no proof ever used it.
Assumption (A5)'s inner pricing regularity (existence, uniqueness and continuity of the price fixed point, assumed, with a $\delta/\sigma_\xi < 1/\phi(0)$ style regularity condition).	SIMPLIFY	Demoted to a theorem under $m_0 \geq 0$: the pricing map is strictly decreasing in P wherever $\bar{m} \geq 0$ and crosses the identity exactly once. A5 survives only as its continuity clause, read as a measurably selected family because the flagged information sets are continuum-indexed. Three independent confirmations, including executed counterexamples that produce zero roots and three roots once $m_0 < 0$.
The discount factor δ inside the pricing fixed point $P = \delta[(1-p)\hat{V} + p(P + \bar{m})]$, and the sensitivity analysis around it.	DROP	v4 prices are undiscounted conditional expectations $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$. The regularity work δ was doing is done instead by $m_0 \geq 0$.
The one-round pooled posterior: $\pi(X, 0)$ in closed form from a single order-flow draw $X \in \{-2, -1, 0, 1\}$ with $p_0 = 1 - \kappa$, $p_1 = \kappa/2$ and the action weights $(\omega_E, \omega_H, \omega_Q)$; Hold and Quiet Voice pooling because both trade $q = 0$.	DROP	Replaced by a pooled history $\mathcal{H}_H^P$ over $H + 1$ business days with plans, Γ -coarsened increments and a flag that can land mid-calendar. The closed-form posteriors do not survive the calendar. Consequential: L3 shows the manuscript's own no-disclosure structure (informed mark $\bar{z}$) yields a <i>four-atom</i> pooled law, two of whose atoms move with κ , so it lies outside $A(\tau)$ — the one-round posterior is not merely superseded, it is a named example of the representation failing.
The hump headline: $\Delta^{\min}(\kappa)$ single-peaked in liquidity, as the conditional-hump proposition and the paper's first main result.	DROP	Not claimed anywhere in v4; the card states outright that it does not claim the hump survives. The v4 headline is the partition and the attenuation factorisation $\mathcal{S} = (1 - \Omega)\mathcal{S}_P$, not a shape in κ . Note the hump has not been <i>refuted</i> either: the enumerated two-round M_P is itself hump-shaped in κ at baseline (peak near 0.55) — it is simply no longer the result being sold, and its role now is to break $A(\tau)$'s orientation clause.
The unconditional attenuation claim: $ \partial\Delta^{\text{act}}/\partial\kappa $ decreasing in ω_P , asserted as a monotone consequence of shifting mass to the disclosed cell, with no composition term and no margin distinction.	DROP	Split and conditioned. The weight effect is kept and proved ($W_T \leq 1$, W_τ likewise), but the composition effect is a separate ratio: T1(B) gets threshold attenuation <i>unconditionally</i> because <i>both</i> W_τ and C_τ lie in $[0, 1]$; T1(C) makes the window margin an iff , $W_T C_T \leq 1$, with C_T unsigned — and the O-1 numbers put three of four calibrated nodes on the <i>failing</i> side. An unconditional window sign is not available.

draft_v2 object	Verdict	What carries over, what changes
Equilibrium uniqueness as a numerical regularity (a 30-seed multistart converging to one fixed point per κ , labelled a regularity rather than a theorem).	DROP	v4 does not claim uniqueness at all, as a theorem or as a regularity. The multistart survives only as an existence diagnostic in the check scripts, and it does not always clear its own binding criterion (Section 10, P1).
The welfare apparatus: transfer netting and the present-value form of total surplus, the liquidity planner's wedge κ^* versus $\kappa^\dagger$, the disclosure-threshold planner, the first-best benchmark.	DROP	Out of scope by the card's own statement: no welfare, no optimal rule design. Note this is also why a bare W is unavailable as a symbol — it was total surplus there, and it is the weight-effect ratio W_τ, W_T here.

10 Evidence

The check-script inventory

Eight scripts, each with committed JSON output, in `quality_reports/fixes/`. Verdicts are as the JSON reports them, failures included. “Wiring” and “substantive” are the design's own classification: a wiring check runs both sides of an identity through the same enumeration and is therefore *not* evidence for the result, only for the code.

Script	Verdict	What it establishes, and what it does not
t2_d1_check	PASS 9 checks, 0 FAIL, 1 vacuous	Clock equivalence by three independent routes (date-by-date scan, one evaluation at $H - T$, closed-form crossing date) agreeing to exactly 0; partition exclusivity and exhaustion at 0 residual; the $T = H$ corner; robustness at $H = 12$. 50 nodes. <i>Vacuous</i> : the Q^F window-monotonicity check, because $\max Q^F = 0$ at the implemented calibration — the flagged round carries no residual order there, so the prediction has nothing to bite on.
t2_l1_check	PASS 4 checks, 0 FAIL	Decomposition residual $\Delta^{\text{act}} - (\Omega M_F + (1 - \Omega)M_P)$ at 0, both degenerate endpoints, and the non-evaluability of the null-cell average. <i>All four are classified WIRING by the design</i> : both sides run through the same enumeration, so the residual is machine noise by construction and is not evidence for L1 .
t2_l2_check	FAIL (1 of 5)	L2's own content passes exactly: the flagged range of M_F over 19 κ values is 0 and its κ -derivatives are 0 — the flagged path never touches the κ -dependent array. Two placebos confirm the pooled side does move. The failure is the fifth check , an <i>ancillary</i> $A(\tau)$ -orientation placebo: see Section 6, fact 2. It is a failed hypothesis about $A(\tau)$ on the enumerated pooled law, not a failure of L2.

Script	Verdict	What it establishes, and what it does not
t2_l3_check	PASS 10 checks, 0 FAIL	The derivative identity, the mean-value form $C_h = \frac{1}{4}\bar{\pi}^2 h''(\zeta)$, the quadratic rate as $\bar{\pi} \downarrow 10^{-4}$, the affine-kernel zero-chord case, and three failure witnesses (a tent kernel with no root, the four-atom Example B gap, an unbounded- h'' witness). <i>Scope</i> : it runs on the three- and four-atom analytic laws and the standalone chord module — it does <i>not</i> enumerate the two-round model, so it verifies L3's mathematics, not $A(\tau)$'s applicability.
t2_l4_check	PASS 10 checks, 0 FAIL	All three sign legs with zero violations over 16 tightening steps (Section 6, fact 3); the exact reclassification identity at 10^{-16} ; flatness in κ at exactly 0 over 71 nodes; the τ -grid spanning a ninefold range in ω_a ; L3's quadratic corollary within 2.1%. Prediction 5 (the size of the A'_κ channel) is <i>reported, not gated</i> , by the request's own instruction: median absolute residual 0.0352, max 0.1213.
t2_p1_check	FAIL (1 of 10)	Nine substantive checks pass: inner-root single crossing and transversality, the flagged family single-valued with the right slope, sequential optimality of the flagged component, both cells on path, the threshold reformulation, monotonicity of Ω in (τ, T) , and <i>existence at the core nodes</i> under the full 30-seed multistart. The failure is the grid sweep : 23 of 27 policy nodes met the binding payoff-scale criterion (10^{-9}) and 4 did not, at best payoff scales 1.5×10^{-3} , 1.1×10^{-3} , 4.0×10^{-4} , 3.1×10^{-4} — all four at extreme liquidity $\kappa \in \{0.15, 0.85\}$, at $\tau \in \{0.05, 0.075\}$ and $T \in \{5, 1\}$ (the independent re-run corrected an earlier gloss here: these are <i>not</i> zero-engagement-share or $T=H$ corner nodes — every actual corner row converged, and the recorded pooled prior engagement share is 0.102). Seed coverage is <i>ruled out</i> as the cause: the check re-ran all four nodes at the full 30 seeds and found identical best residuals, with most seeds landing on the same cutoff vector. All 27 nodes do converge on the cutoff-scale criterion. The cause is undiagnosed — candidates are a failure of A3's single crossing at those κ extremes (the adjacent-plan payoff gap is a sawtooth on this menu, and single crossing is a calibration fact, not structural) or a genuine boundary non-existence outside P1's hypothesis set; either way P1's theorem is untouched (its hypotheses include A3) and the four nodes are named, not smoothed.
t2_t1_check	FAIL (1 of 9)	The factorisation $\mathcal{S} = (1 - \Omega)\mathcal{S}_P$ holds pointwise to 1.2×10^{-16} and in total variation to 2.2×10^{-19} ; Ω is flat in κ at 0 residual; the threshold margin, the window margin, the O-1 benchmark and the composition factors all pass; $H = 12$ robustness passes. The failure is block 3 , the chord-magnitude route: see Section 6, fact 2. One check is <i>vacuous</i> (the local form, which needs a smooth window interpolation the integer calendar does not provide).

Script	Verdict	What it establishes, and what it does not
t2_c1_region_check	PASS 12 checks, 0 FAIL, region nonempty	80 nodes ($8 \kappa \times 5 \tau \times 2 T$), 2.9 hours of wall time. Two gated verdicts, both PASS with 0 violations: the bound contains the four-corner re-solve remainder at the 8 validation nodes (max ratio 0.511), and it contains the implicit-function remainder at all 80 (max ratio 1.0). 18 nodes certified , all at $T = 5$ and the upper three τ percentiles; $\eta_r \in [0.0595, 1.7227]$, median 0.3467; $L_{\mathcal{R}} \in [0.264, 0.501]$ with no node at $L \geq 1$. <i>Failure attribution</i> , recorded not hidden: 56 nodes have $g_r^{PE} = 0$ to 10^{-10} because the implemented legal clock quantises the crossing date, so $\Omega(\tau)$ is locally constant and the cross-derivative vanishes identically there; 4 are positive but dominated; 2 are negative; 40 nodes (all $T = 10$) are degenerate on flagged mass below 0.01. The sharper, non-inversion-free bound would certify 22 rather than 18 — reported as a measure of the Neumann step’s price, and certifying nothing. No tolerance was weakened to make the region nonempty.

Provenance caveat. Seven of the eight scripts record the card stamp *2026-08-20* in their provenance block; only `t2_c1_region_check` was run against the 2026-08-21 stamp. The card changes between those stamps are hypothesis-explicitness moves, not model changes, but the mismatch is on the record and is not papered over here.

The `numerical_v4` package: smoke facts

The implementation is `numerical_v4/` (modules `params`, `menu`, `policy`, `flagged`, `pooled`, `premium`, `solver`, `smoke`), with committed output in `numerical_v4/smoke_output.txt`. At the smoke configuration $H = 10$, $T = 5$, $M = 2$ order marks, $J = 3$ plans, $b_0 = 0.03$, $\bar{b} = 0.1$, seed $\tau = 0.05$ (the statutory 13D 5%):

- **Enumeration gate:** 4,194,304 histories, 826,686 feasible, $N_\theta = 12$; feasible $\times N_\theta = 9.92 \times 10^6$ against a gate limit of 10^8 . PASS. One evaluation takes 0.372 s; the whole smoke runs in 64.8 s.
- **τ freezing:** the seed equilibrium is $k = (1.31246481, 1.41888743)$ with $|k - \mathcal{T}(k)| = 1.34 \times 10^{-11}$; the median Voice $b^*(s)$ is 0.09076406, and τ is frozen there at every node.
- **Baseline equilibrium** ($\kappa = 0.5$): $k = (1.2405757283, 1.5310222869)$; outer residual 2.60×10^{-11} on the diagnostic cutoff scale and *exactly zero* on the binding payoff scale.
- **Masses:** $\Omega = 13.84\%$, $\mathbb{P}(a = 1) = 22.63\%$, $\omega_a = 61.15\%$, $\bar{\pi}_{\text{pr}} = 10.21\%$. *Note* $\omega_a = 61\%$ here is a property of the calibration, not an estimate: the card claims no empirical value for ω_a , and the empirical anchor for it is absent from the literature.
- **Premium objects:** $M_F = 0.5528$ pp, $M_P = 0.2228$ pp, $\Delta^{\text{act}} = 0.2685$ pp; L1’s decomposition residual is 0 (wiring).
- **Price path:** mean run-up $R = 389.18$ bp, mean filing jump $J = 2590.52$ bp, D1 identity residual 0 against a 10^{-12} tolerance.
- **A7 certificate:** passes — minimum $|db^*/ds|$ on the flagged set is 2.20×10^{-4} (gate 10^{-8}), no flat sub-intervals, no (B^F, Q^F) collisions, no cross-plan collisions, 10 flagged atoms of 19.
- **Numerical hygiene:** pooled mass error 2.2×10^{-16} ; maximum inner price residual 2.2×10^{-16} ; maximum b^* inversion error 5.9×10^{-14} ; GL-20 versus GL-40 quadrature on M_F differing by 2.6×10^{-18} ; **zero multiple-root nodes** anywhere, which is the structural consequence of $m_0 \geq 0$.

- **L2 verdict, substantive:** the range of M_F across the frozen-policy κ sweep is *exactly* 0.
- **The shape that breaks $A(\tau)$:** M_P is hump-shaped in κ with its peak near $\kappa = 0.55$, so $\mathcal{S}_P = |\partial_\kappa M_P|$ is V-shaped about that peak, ranging from 4.99×10^{-3} to 2.66×10^{-1} premium pp per unit κ ; at the peak $\mathcal{S} = (1 - \Omega)\mathcal{S}_P = 6.37 \times 10^{-2}$.
- **Chord module:** $C_h \leq 0$ at every $\bar{\pi}$ tested, and $|C_h|/\bar{\pi}^2$ settles at ≈ 0.2219 as $\bar{\pi}$ falls from 10^{-1} to 10^{-4} — L3’s quadratic rate, on the standalone route.
- **One structural caveat:** $\max Q^F = 0$ at this calibration. The flagged round exists in the timing and in the code, but carries no residual order here, which is why the Q^F monotonicity check is vacuous.

Sources, all in the v4-theory worktree: `research/model_v4/MODEL_CARD.md` (stamp 2026-08-21, post-ledger regeneration) for Sections 1–6 and 7; `research/model_v4/LABEL_LEDGER.md` for the labels; `research/model_v4/proofs/` and `research/model_v4/rederive/` for the two-pass evidence chains; `research/model_v4/HANDOFF_sign.md` and `quality_reports/fixes/t1_o1_rerun_check.json` for the O-1 numbers; `quality_reports/fixes/t2_*_check.json` for the check inventory; `numerical_v4/smoke_output.txt` for the smoke facts.